WEEK 4 – END-TO-END SALES DATA ANALYSIS

Internship: Data Science with Python
Project Title: Sales Performance Analysis
Tools Used: Python, Pandas, Matplotlib, Seaborn and Power BI

1. Introduction

The objective of this project is to analyse business sales data, identify sales trends, understand product performance and generate useful business insights.

2. Data Cleaning

- Loaded the Excel dataset using Pandas.
- Checked dataset shape and column names.
- Identified 53 missing values in the Discount Band column.
- Replaced missing values with "Unknown".
- Checked duplicate records.
- Explored the dataset using descriptive statistics.

3. Data Analysis

Performed the following analyses using Python and Pandas:

- Sales analysis by product.
- Profit analysis by product.
- Country-wise sales analysis.
- Yearly sales analysis.

4. Key Findings

- **Total Sales:** 118.73 million
- **Total Profit:** 16.89 million
- **Total Units Sold:** 1,125,806
- **Best-selling Product:** Paseo
- **Most Profitable Product:** Paseo
- **Top Sales Country:** United States of America
- **Total Products:** 6
- **Total Countries:** 5

5. Data Visualization

Created charts using Python to represent:

1. Total Sales by Country.
2. Total Sales by Product.
3. Total Profit by Product.
4. Yearly Sales Analysis.

Also created a Power BI dashboard containing sales and profit cards, product and country charts, and interactive Year and Country filters.

6. Business Recommendations

1. Focus on promoting the Paseo product because it generates the highest sales and profit.
2. Investigate opportunities to improve sales in Mexico.
3. Study successful sales strategies in the United States.
4. Monitor product performance and profitability regularly.
5. Use interactive dashboards to support better business decisions.

7. Conclusion

This project helped me understand the complete data analysis process, including data cleaning, exploration, analysis and visualization. Python and Power BI helped convert raw sales data into meaningful business insights that can support decision-making.

8. Learning Outcomes

- Learned data cleaning using Pandas.

- Practised grouping and analysing datasets.
- Created visualizations using Matplotlib.
- Developed an interactive dashboard using Power BI.
- Improved my understanding of business data analysis.